

Lesson 41: Integration & Antiderivatives — Practice Worksheet

Name: _____ Date: _____

Attempt every question. Show your working. The sections increase in difficulty.

Section 1 — Warm-up (foundation)

1. Find $\int x^3 dx$.
2. Find $\int 4x dx$.
3. Find $\int (2x + 1) dx$.
4. Find $\int x^5 dx$.
5. Find $\int 6 dx$.
6. What must you always add for an indefinite integral?

Section 2 — Core practice

7. Find $\int (6x^2 + 2) dx$.
8. Find $\int (3x^2 - 4x + 5) dx$.
9. Find $\int 6x^2 dx$.
10. Find $\int (x - 3) dx$.
11. Find $\int (4x^3 - 1) dx$.

12. Find $\int 10x^4 \, dx$.

Section 3 — Challenge & exam-style

13. A curve has $dy/dx = 4x$ and passes through $(1, 7)$. Find y .

14. A curve has $dy/dx = 6x^2 - 2$ and passes through $(1, 3)$. Find y .

15. Given $dy/dx = 3x^2$ and the curve passes through $(2, 10)$, find y .

16. Find $\int (x^2 + 1)^2 \, dx$ (expand first).